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1  /* =====
2  1. DEPOSIT_MONEY
3  Covers: Cash Deposit, ATM Deposit, Cheque Deposit, Interest Credit
4  ATM transactions are additionally recorded in ATMTransactions
5  =====*/
6  CREATE OR ALTER PROCEDURE Deposit_Money
7  (
8      @AccountId INT,
9      @DepositAmount DECIMAL(18,2),
10     @Category VARCHAR(30), --(Cash Deposit | ATM Deposit |
11                                | Cheque Deposit | Interest Credit)
12     @Channel VARCHAR(20), --(Branch | ATM | Internet Banking
13                                | Mobile Banking | System)
14     @ATMID INT = NULL
15 )
16 AS
17 BEGIN
18     SET NOCOUNT ON;
19
20     BEGIN TRY
21         BEGIN TRANSACTION;
22
23         -- 1. Validate deposit amount
24         IF @DepositAmount <= 0
25         BEGIN
26             THROW 50001, 'Deposit amount must be greater than zero.', 1;
27         END
28
29         -- 2. Validate deposit category
30         IF @Category NOT IN
31         (
32             'Cash Deposit',
33             'ATM Deposit',
34             'Cheque Deposit',
35             'Interest Credit'
36         )
37         BEGIN
38             THROW 50003, 'Invalid deposit category.', 1;
39         END
40
41         -- 3. Validate account
42         IF NOT EXISTS
43         (
44             SELECT 1
45             FROM Accounts
46             WHERE AccountID = @AccountId
47                   AND AccountStatus = 'Active'
48         )
49         BEGIN
50             THROW 50002, 'Account does not exist or is not active.', 1;
51         END
52
53         -- 4. Validate Category + Payment Channel
54         IF
55         (
56             (@Category = 'Cash Deposit' AND @Channel NOT IN ('Branch', 'ATM'))
57             OR
58             (@Category = 'ATM Deposit' AND @Channel <> 'ATM')
59             OR
60             (@Category = 'Cheque Deposit' AND @Channel NOT IN ('Branch', 'ATM',
61                                                                'Internet Banking', 'Mobile Banking'))
62             OR
63             (@Category = 'Interest Credit' AND @Channel <> 'System')
64         )
65         BEGIN
66             THROW 50004, 'Invalid Category and Payment Channel combination.', 1;
67         END
68
69         -- 5. Validate ATM details when transaction is through ATM
70         IF @Channel = 'ATM'
71         BEGIN
72             IF @ATMID IS NULL
73             BEGIN
74                 THROW 50005, 'ATMID is required for ATM transactions.', 1;
75             END
76
77             IF NOT EXISTS
78             (
79                 SELECT 1
80                 FROM ATM
81                 WHERE ATMID = @ATMID
82                       AND Status = 'Active'
83             )
84             BEGIN
85                 THROW 50006, 'ATM does not exist or is not active.', 1;
86             END
87         END
88
89         -- 6. ATMID must be NULL for non-ATM transactions
90         IF @Channel <> 'ATM' AND @ATMID IS NOT NULL
91         BEGIN
92             THROW 50007, 'ATMID must be NULL for non-ATM transactions.', 1;
93         END
94
95         -- 7. Get account opening balance
96         DECLARE @OpeningBalance DECIMAL(18,2);
97
98         SELECT @OpeningBalance = CurrentBalance
99         FROM Accounts
100        WHERE AccountID = @AccountId;
101
102         -- 8. If ATM transaction, validate ATM cash capacity
103         DECLARE @ATMOpeningBalance DECIMAL(18,2);
104         DECLARE @ATMClosingBalance DECIMAL(18,2);
105         DECLARE @CashCapacity DECIMAL(18,2);
106
107         IF @Channel = 'ATM'
108         BEGIN
109             SELECT
110                 @ATMOpeningBalance = CashBalance,
111                 @CashCapacity = CashCapacity
112             FROM ATM
113             WHERE ATMID = @ATMID;
114
115             IF @ATMOpeningBalance + @DepositAmount > @CashCapacity

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-- 9. Update customer account balance
UPDATE Accounts
SET
    CurrentBalance = CurrentBalance + @DepositAmount,
    AvailableBalance = AvailableBalance + @DepositAmount,
    LastTransactionDate = SYSDATETIME(),
    ModifiedDate = SYSDATETIME()
WHERE AccountID = @AccountId;

-- 10. Insert financial transaction
INSERT INTO Transactions
(
    TransactionReferenceNumber,
    TransferReferenceNumber,
    AccountID,
    TransactionType,
    TransactionCategory,
    PaymentChannel,
    Amount,
    Charges,
    TaxAmount,
    OpeningBalance,
    ClosingBalance,
    CounterpartyAccountNumber,
    CounterpartyName,
    CounterpartyBank,
    TransactionStatus,
    Remarks,
    InitiatedBy,
    ApprovedBy,
    TransactionDate,
    CreatedDate
)
VALUES
(
    'DEP-'
    + CONVERT(VARCHAR(8), CAST(SYSDATETIME() AS DATE), 112)
    + '-'
    + RIGHT('000000' + CAST(NEXT VALUE FOR TransactionReferenceSeq
                           AS VARCHAR(6)),6),
    NULL,
    @AccountId,
    'Credit',
    @Category,
    @Channel,
    @DepositAmount,
    0.00,
    0.00,
    @OpeningBalance,
    @OpeningBalance + @DepositAmount,
    NULL,
    NULL,
    NULL,
    'Success',
    'Deposit made via ' + @Channel,
    NULL,
    NULL,
    SYSDATETIME(),
    SYSDATETIME()
);

-- 11. Capture generated TransactionID
DECLARE @TransactionID INT;
SET @TransactionID = SCOPE_IDENTITY();

-- 12. ATM-specific processing
IF @Channel = 'ATM'
BEGIN
    -- 12.1 Update physical ATM cash
    UPDATE ATM
    SET
        CashBalance = CashBalance + @DepositAmount,
        LastCashLoadDate = SYSDATETIME()
    WHERE ATMID = @ATMID;

    -- 12.2 Get remaining ATM cash
    SELECT @ATMClosingBalance = CashBalance
    FROM ATM
    WHERE ATMID = @ATMID;

    -- 12.3 Record ATM transaction
    INSERT INTO ATMTransactions
    (
        ATHID,
        AccountID,
        TransactionID,
        TransactionType,
        Amount,
        RemainingATMBalance,
        TransactionDate
    )
    VALUES
    (
        @ATMID,
        @AccountId,
        @TransactionID,
        'Deposit',
        @DepositAmount,
        @ATMClosingBalance,
        SYSDATETIME()
    );
END;

-- 13. Commit complete operation
COMMIT TRANSACTION;

-- 14. Return result
SELECT
    @AccountId AS AccountID,
    @OpeningBalance AS OpeningBalance,
    @DepositAmount AS DepositAmount,
    (@OpeningBalance + @DepositAmount) AS ClosingBalance,
    @TransactionID AS TransactionID,
    @ATMID AS ATMID,
    'Deposit successful.' AS Message;
END TRY
BEGIN CATCH
    -- Roll back all changes if any error occurs
    IF @TRANSCOUNT > 0
        ROLLBACK TRANSACTION;
    THROW;

```

